MEC 316 NOTES P1

Candidate's Number..........

VOL. 2 Question CAD/CAM

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Numerical Control (NC) – This is the operation of machine tools and certain other processing machines by series of coded instruction, w/c consist of digits, or numbers and other symbols, including letters. NC is a system that uses prerecorded information prepared from numerical data to control a M/C tool or other machining process.

NC describes the control of machines movements and various other functions by instructions expressed as a series of numbers & initiated via electronic control system.

COMPUTER NUMERICAL CONTROL (CNC)

This is the term used when a computer is included in the control of the machine. Thus, CNC means that a computer controls the movement of a machine by computer aided programs that are made up of letters, symbols and numbers. Most of the common types are: Milling M/C, drill presses, grinders, turret punches, welding M/C, and lathe M/C.

– PROGRAMS: –

CNC M/C contain step-by-step instruction to tell the M/C every movement of the tool or work piece, the cutting speed and the feeds of each machining operation, the start and end of each machining operation, and the start and end of coolant flow.

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* Once a program is properly written for CNC M/C, parts are produced automatically without the skilled machinist to run the M/C *

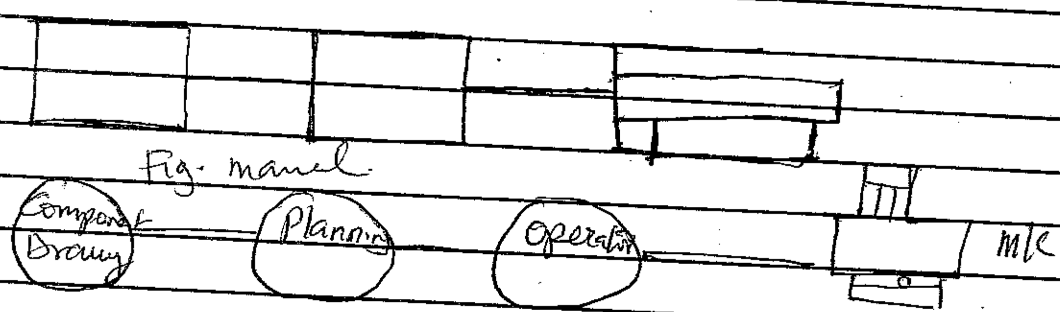

Fig. manual

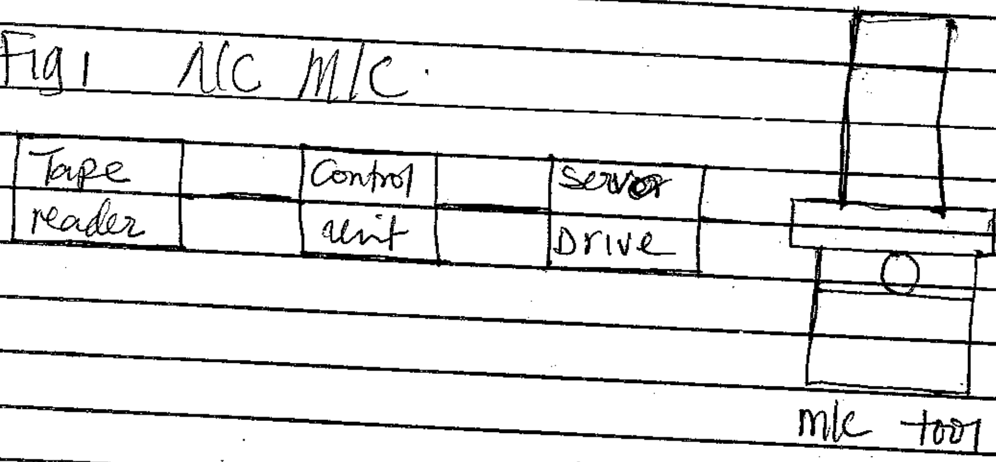

Fig 1 NC M/C

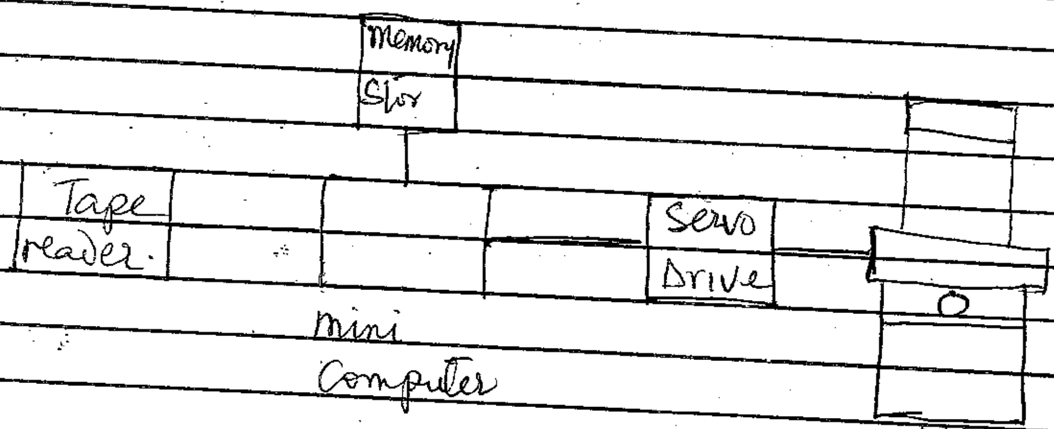

fig 2 - CNC

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NC M/C can be used when

1. The tooling cost is high compared to the machining cost by conventional method.
2. The set up time is large in conventional M/C.
3. Frequent changes in tooling & machining setting are required.
4. Parts are produced intermittently.
5. Complex-shaped components are needed.
6. Expensive parts where human errors are costly.
7. Design changes are frequently.
8. 100% inspection is required.

ADVANTAGES OF NC OVER CONVENTIONAL M/C.

1. Greater flexibility – wide variety of operation can be carried through tape program.
2. Elimination of templates, models, jigs and fixtures
3. ~~Asie~~ Easier set-ups – operator don't need to set table limits or dogs.
4. Reduced machining time – use wide range of speeds than conventional m/c.
5. Greater accuracy and uniformity – no operator error.
6. Greater safety – operator not so close to machine.
7. Conversion to metric system; easy to accept either inch or metric input.

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DISADVANTAGES OF NC TO CONVENTIONAL

1. NC follows programmed instruction that can lead to M/C distruction if not properly prepared.

2. NC cannot add any extra machining capabilities to the M/C tool if no added in the program.

3. NC M/C cost more than conventional M/C of the same capacity.

4. Higher skilled personel needs to operate NC b/cos of the high tech. involved.

5 Special Training for personel in both soft & hard ware is neccessary

6. NC requires higher investments in terms of wages of skilled worker & expensive spare parts.

ADVANTAGE OF CNC OVER NC

* NC M/Cs are not connected to a computer, thus lacks memory capabilities to run a canned cycle.

* NC lacks memory to calculate arcs and curves instaltenously.

* CNC have added feature of graphics

CAD/CAM Cont. from (P4)

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Simulation: This feature allows the programmer to watch visual display of how the CNC M/c will machine the parts according to the programme that was created.

PRINCIPLE OF OPERATION:

The principle of operation of the N/C m/c can be explained with the help of a vertical milling M/C. To carry out an operation like end milling the spindle head is to be positioned in Z-axis and the table in X & Y coordinate axes. See diagram bellow.

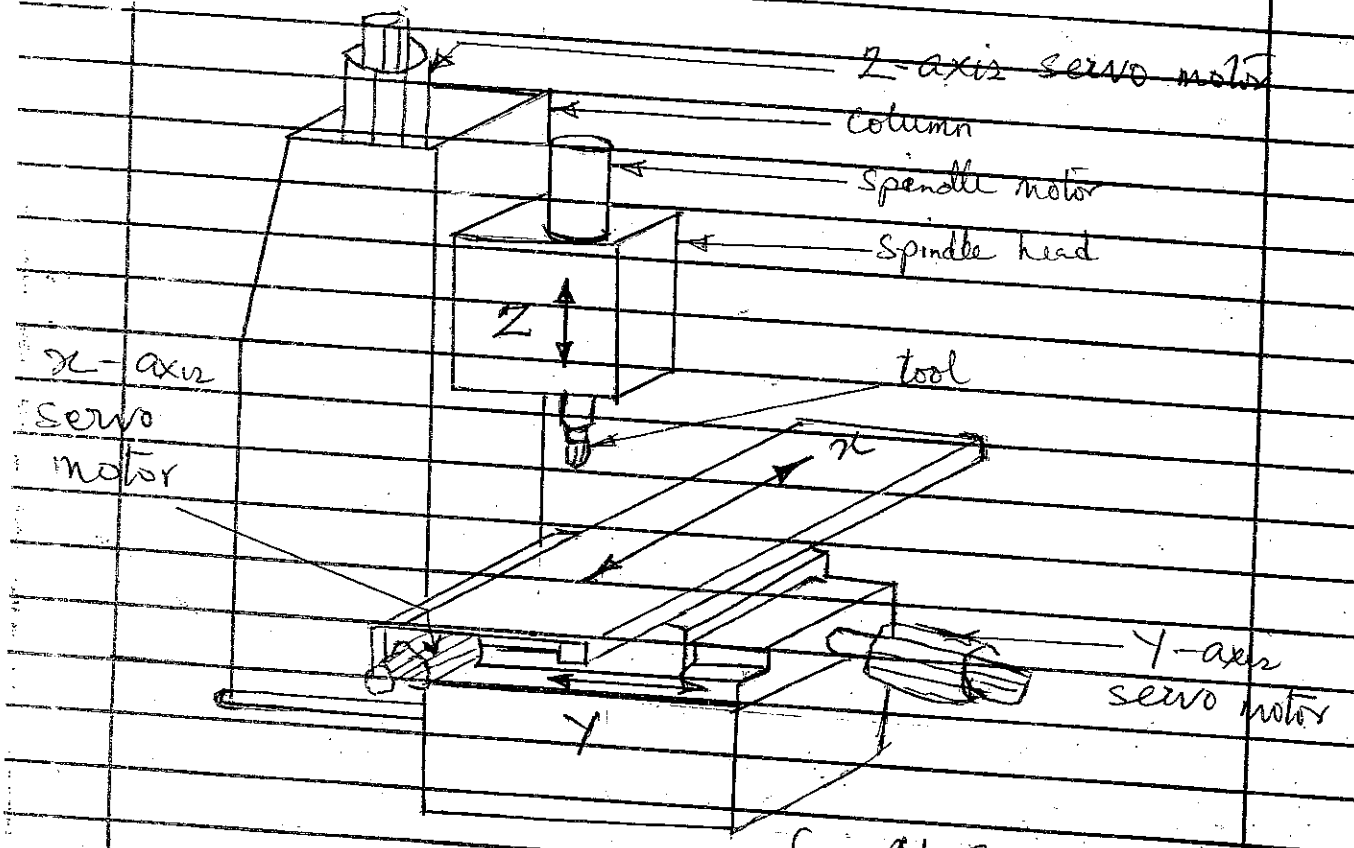

fig. N.C. m/c

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The feed movement is to be realised by the individual or simultaneous movement of X & Y axes. Thus, the milling M/C requires three slide movements refered to as axes feed drives.

A special feature of a CNC M/C is that, a separate motor called a servomotor individually drives each axis.

TYPES OF CNC MACHINES:

1. Machine Centre
   · Horizontal, vertical & universal types
2. CNC lathe
3. CNC turning centres
4. Turn mill centre
5. CNC milling/drilling/plana M/Cs
6. Gear Hobbing M/Cs
7. Gear Shaping M/C
8. Wire cut EDM/ECDM
9. Tube bending
10. Laser/arc/plasma cutting M/C
11. Electric beam welding
12. Coordinate measuring M/C
13. Grinding M/Cs.
   · Surface grinder, · cylinderical grinder and centre less grinder.
14. Tool & Cutter grinder
15. CNC Boring & jig boring M/C
16. PCB drilling M/C
17. press brakes
18. CNC Guillotines

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19 CNC Transfer lines SPM's
20 Electromechanical milling M/C
21 Abrasive ~~water~~ water jet cutting M/C
22 Flow forming M/C
23 Roll forming "
24 Turret punch press

CNC PROGRAMMING

CNC machines are programmed using a standardized list of codes that were developed by the electronic industries Association (EIA). The codes ~~are~~ include preparatory functions or 'G' codes. Some of these ~~codes~~ functions include cutting an arc in a clockwise directions (G02 - circular interpolation) and telling the machine that the dimension will be given in millimeters (G71 - metric programming). The Electronic industries Association codes are given as fellow.

CODES	Description Preparatory fns
G00	Denotes rapid traverse for point-to-point positioning
G01	Linear interpolation.
G02	Circular interpolation clockwise
G03	Circular interpolation anticlockwise.
G04	Dwell
G05-07	Unassigned
G08	Acceleration at smooth rate
G09	Deceleration at smooth rate

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G10 – 16	unassigned
G13 – 16	~~AX~~ Axis Selection Codes.
G17XY	Plane selection
G18 ~~Z~~X	Plane Selection
G19 YZ	Plane Selection.
G20-32	Un assigned
G33	Thread Cutting, Constant lead.
G34	Thread cutting, increased lead.
G35	Thread cutting decreased lead
G36-39	unassigned
G40	Cutting diameter Compensation ~~Ca~~ Cancels.
G41	Cutting diameter Compensation left
G42	Cutting diameter Compesation right
G43	cutting diameter Compensation inside Corner (used to adjust for differences in programmed & actual Cutter size)
G44	Cutter diameter Compensation outside Corner (used to adjust for differences in programmed & actual Cutter size)
G45-49	unassigned
G50-59	used with adaptive Controls.
G60-69	Un assigned.
G70	Inch programming
G71	metric programming
G72	Three-dimensional Circular interpolation clockwise
G73	Three-dimensional circular interpolation Counterclockwise

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Code	Description
G74 - ~~Multi~~	Multi-quandrant circular interpolations cancel.
G75 -	~~multiple~~ multi-quadrant circular interpolation.
G76-79	Unassigned
G80	cycle cancel
G81	Drill circle
G82	Drill circle with duell
G83	Intermitted or deep hole drilling circle
G84	Tapping circle.
G85-89	Boring cycles.
G90	Absolute positions.
G91	Incremental positioning.
G92	Register preload code
G93	Inverse time feed rate.
G94	Inches (milimeters) per minute feed rate
G95	Inches (millimeters) per revolution feed rate
G96	Unassigned
G97	Revolutions per minute spendle speed.
G98-99.	Un assigned.

Miscellaneous Functions

M00	program stop
M01	optional (planned) stop
M02	End of program
M03	spindle on clockwise
M04	Spindle counter clockwise.
M05	Spindle off
M06	Tool change
M07	Coolant on (flood)
M08	Coolant on (mist)

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Code	Discription
M09	Coolant Stop
M10	Automatic clamp
M11	Automatic Unclamp
M12	Synchronize Multiple axes
M13	Spindle Clockwise & coolant on
M14	Spindle Counterclockwise & coolant on
M15	Rapid Motion positive direction
M16	Rapid Motion -ve direction
M17-18	Unassigned
M19	Spindle orient & Stop
M20-29	Unassigned
M30	End of Tape (will rewind tape automatically)
M31	Interlock ~~by~~ bypass
M32-39	Unassigned
M40-46	Gear changes if used otherwise Unassigned
M47	Continues program execution from start
M48	Cancel M47
M49	Deactivate manual speed or feed override
M50-57	~~Cancel~~ Unassigned
M58	Cancel M59
M59	RPM hold
M60-~~89~~	Unassigned

OTHER ADRESSES

A	Rotary Motion about the X-axis
B	" " " " Y-axis
C	" " " " Z-axis
D	Angular dimension around special axis (also used for a third feed function
E	Angular dimension around special axis or special feed function

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H – Unassigned.

I – X-axis arc center point.

J – Y-axis arc center point.

K – Z-axis arc center point.

L – Unassigned.

O – Used on some controllers in place of N address of sequence numbers

P – Special rapid traverse code or third axis parralled to the X-axis.

Q – Special rapid traverse code or third axis parallel to the Y-axis

R – Special rapid traverse code or third axis parallell to the Z-axis. (also used for radius designation)

U – Secondary axis Parallel to X-axis

V – " " " " Y-axis

W – " " " " Z- "

CAD/CAM SYSTEM:

Cad/cam – system allows a single person to draw a part three dimensionally, select a tool for machining it, create tool paths by picking points on the edge of a part, and download it to a machining centre for immediate processing.

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Coordinates Axis & Motion

Most of the program comprises of points & coordinates which refer distance to axes. These Machine determine the distances along the x, y, & z coordinates and use these coordinates for tool motion or positioning tools on the work or for references.

There are nine standard axis used in CNC machining

a. Primary Linear (straight line) movement X, Y & Z. These axes are used to define three-space (3-dimensional) envelop lies at 90° to each other and thus called an "orthogonal axis set"